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IE4512: Engineering Economy



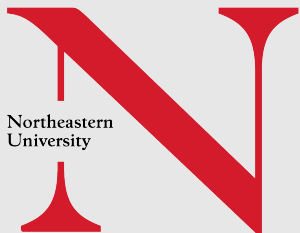
Prof. Mohammad Dehghani

"The most valuable of all capital is that invested in human beings; and of that capital the most precious part is ..." [Google it]

- Alfred Marshall - a British economist

"The best investment you can make is in yourself ... the more you learn, the more you'll earn"

- Warren Buffett



Engineering Economy

IE4512

Instructor Info —



Course Info —



TA Info —



Course Overview

This course explores economic modeling and analysis techniques for selecting alternatives from potential solutions to an engineering problem. The course begins with exploring measures of merit and basic methods of economic comparison, such as present worth, annual worth, rate of return, and benefit/cost techniques. Further moving on to examine recent techniques of economic analysis, especially the tools of decision-making. Along with that, the course considers causes of risk and uncertainty, effects of taxes on investment, decision tree analysis, statistical decision techniques, and multiple attributes of decision making through group case studies.

Course Objectives

- Identify the key financial vehicles that effect how companies do business and describe their impact on economic decisions.
- Analyze projects and ventures using standard economic decision-making techniques such as breakeven and sensitivity analysis, risk analysis and uncertainty, decision trees, and other statistical decision techniques problem solving.
- Analyze and Evaluate a variety of complex real -world economic scenarios by applying multiple attributes of decision-making principles.
- Evaluate projects based on dynamic factors such as depreciation and taxes to choose the optimal alternatives.
- Be able to compare competing projects by understanding time value of money principles to cash flows.
- Apply and incorporate decision-making techniques to include economical and non-economical factors into the alternatives selection process.
- Determine the economic equivalence and manipulate cash flows in order to make economic projections.

Reference Books

- Sessions: As per the Itinerary
- Principles of Engineering Economic Analysis, 6th Edition by John A. White, Kenneth E. Case, and David B. Pratt, March 2012 [Amazon](#)
 - ~~Check the bookstore~~ [WileyPLUS](#) with the Wiley-Plus, so there is no need to purchase the hard copy of the text-book

Online textbook with access to WileyPLUS material in Canvas

- Use the digital (online) version of the WileyPLUS text within Canvas, and purchase access online through www.wileyplus.com. You are presented with the option to buy WileyPLUS digital access the first time you click on a link to WileyPLUS material from the IE4512 Canvas course site

Printed textbook with access to WileyPLUS material in Canvas
Office Hrs: Monday - Friday: 11am - 12pm

- Buy the printed textbook from the NU Online bookstore and a WileyPLUS access code will be included! In-person
- ~~DO NOT THROW THIS CODE AWAY~~ feijoo@northeastern.edu

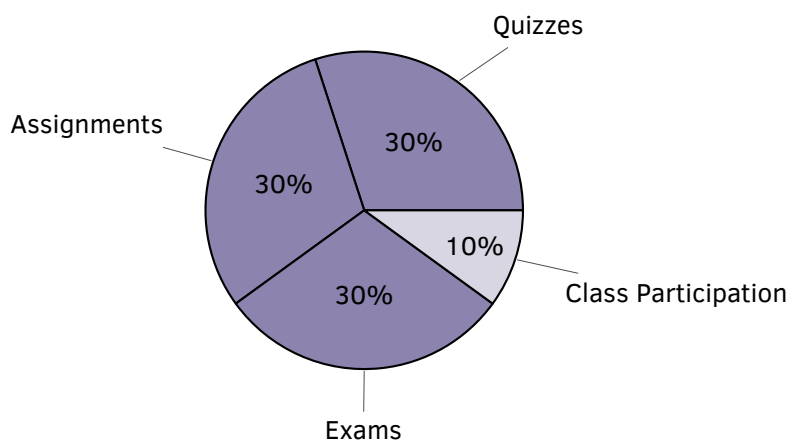
Class Schedule

Please check CANVAS and course itinerary for the daily schedule of the course. If due to unforeseen circumstances the schedule changes, students will be informed prior to the change in schedule.

Topical Outline

Module	Topics	Readings
1	Economic Principles and an Introduction to the Time value of Money (TVOM)	Ch. 1 and Ch. 2.1 to 2.3
2	Analyzing Cash Flows and Series of Cash Flows	Ch. 2.4 to end of Ch.2
3	Borrowing, Lending, and Investing	Ch. 3
4	Planning Horizon, MARR and Present Worth Analysis	Ch. 4 and Ch. 5.1 to 5.5
5	Present, Future and Annual Worth Analysis	Ch. 5.6 to end of Ch. 5, Ch. 6 and 7
6	Rate of Return Analysis	Ch. 8
7	Depreciation Method	Ch. 9
8	After-Tax Economic Analysis	Ch. 10
9	Decision making with uncertainty	Ch. 13.1 to 13.3

Grading Scheme



► Excel Assignments (Individual)

- Up to 9 Excel Assignments will be given to you. Students need to follow the provided instructions/lectures and solve the Excel problems. These problems represent the book examples but require excel skills to solve.
- Assignments submitted have to be labeled as "IE4512 - HW # <Student Name>"
- Late assignments **will not be** graded.
- The worst scored Excel Assignment will be dropped and will not count towards the final grade.

► Quizzes (Individual)

- The Quizzes will be assigned through Canvas. These Quizzes will be given after each module is completed.
- Quizzes are due according to the deadlines listed on canvas. The online system will automatically deduct points for late submissions.
- The WORST Quiz grade will be dropped and will not be counted towards the final grade.
- For each quiz you would have 2 attempts.
- Please refer to the Quiz Excel template provided on canvas under Assignments tab.

► Exams: Mini Exam 1 and Mini Exam 2 (Individual)

- The exams will be take home and will be announced 1-2 weeks in advance.
- Both exams are equally weighted.

► Class Participation & Discussion

- The course is synchronous and attendance is mandatory for students taking the course. To have an effective learning experience, students are encouraged to have continuous interaction with the faculty. A discussion question will be given to each section/break-out rooms to stimulate critical thinking. All students need to actively participate in discussions to articulate their ideas and share their thoughts.
- Performance will be evaluated based on their participation and quality of response.

Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. In fact, you

should audit the correct entry of your grades. No grade corrections will be made after the last week of the semester. Final Grades and Course Letter Grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

This course prepares students to achieve student outcomes 1, 2, 3, 4, and 7.

Student Outcome	1	2	3	4	5	6	7
Emphasis	1	3	1	1	1	-	2

Emphasis

- 3 Strong Support ; 2 Support ; 1 Somewhat support

Emailing Us?

Please use the following format for the subject line of your email to avoid miss communication. This subject line will reduce the confusion.

IE4512 - Subject Line

Make-up Policy

- No makeup opportunity will be given for any homeworks/lab assignment/projects.
- Students unable to attend classes are responsible for obtaining information regarding announcements and lecture content

Class Policy

- Students are expected to attend all class sessions and are responsible for all materials presented in class. Past experience has shown students who attend regularly and interact with instructors/classmates tend to be better overall.
- Laptops: You may need to use your laptop during the class upon the instructor's notes
- Phones/tablets/smart watches: Electronic devices may not be used during the class. If you have class work, social networking or sports-score browsing, that is more important to do, then please skip class. Life is full of such choices.
- Discussion between students for homeworks and projects is permissible. However the ideas / solutions / codes must not be copied. Each individual submission must be an outcome of individual effort

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity [website](#) to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act

(ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center [website](#)

Northeastern University Copyright Statement

This course material is copyrighted, and all rights are reserved by Northeastern University. No part of this course material may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual, or otherwise, with the express prior written permission of the University.

Diversity and Inclusivity Statement

Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the [website](#) for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.

In case of an emergency, please call 911.

Please visit the [website](#) for a complete list of reporting options and resources both on- and off-campus